2-WAY CROSSOVER

2x 22W-8534G00 & 4526ND

Complete Simulation Graphs & Crossover Design

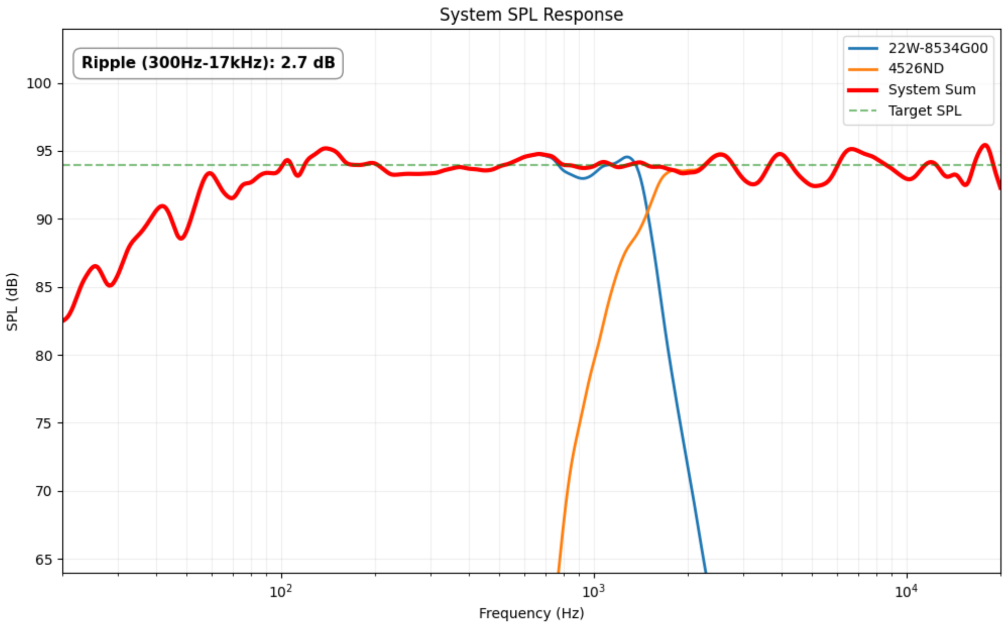
AUDIO440

Contents

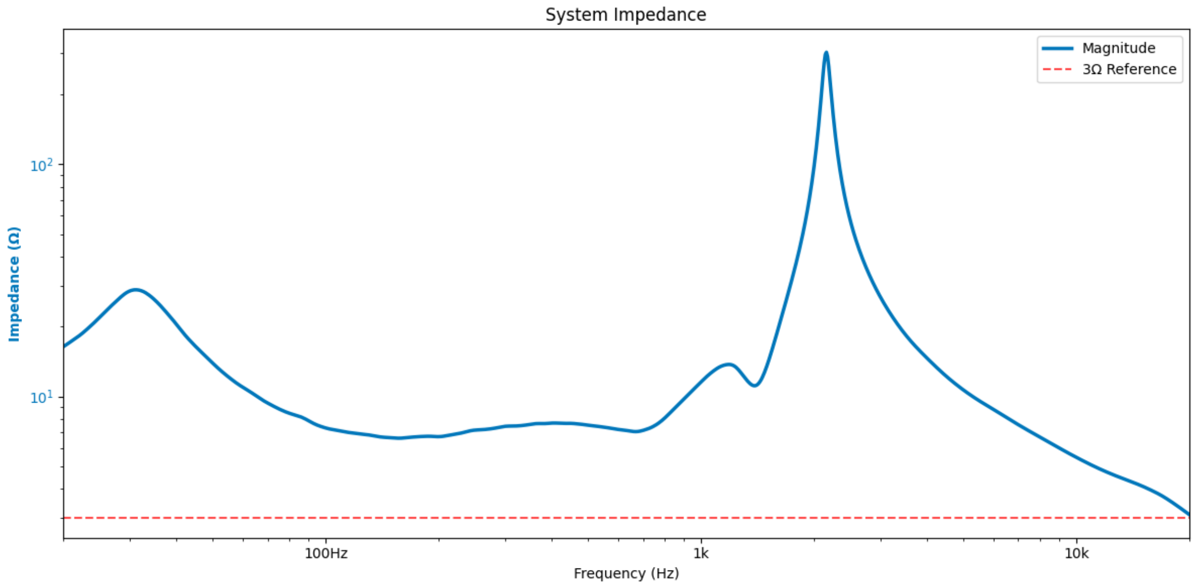
1 Acoustic Simulations

Simulation graphs showing the frequency response of the drivers combined with the crossover. For more information on how to interpret these graphs, please refer to the beginner guide provided with your order.

1.1 On-Axis SPL Response



1.2 Impedance Response



2 Bill of Materials (BOM)

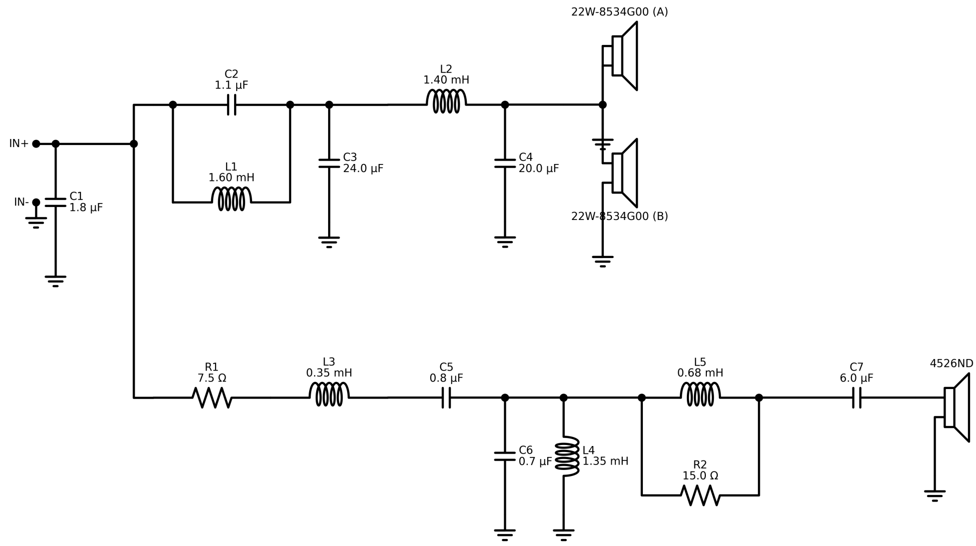
The following table lists all the components required to build one crossover. For a stereo pair, please double the quantities. Note that the provided prices and links are for reference only and may vary over time. Always check the supplier's website for the most up-to-date information.

ID	Component	Value	Price ($/€)	Buy Link
C1	Capacitor	1.8 μF	1.31	Link
C2	Capacitor	1.1 μF	1.49	Link
L1	Inductor	1.6 mH	9.45	Link
C3	Capacitor	24.0 μF	16.10	Link
L2	Inductor	1.4 mH	9.45	Link
C4	Capacitor	20.0 μF	5.09	Link
R1	Resistor	7.5 Ω	0.99	Link
L3	Inductor	0.35 mH	8.45	Link
C5	Capacitor	0.82 μF	1.49	Link
C6	Capacitor	0.68 μF	0.59	Link
L4	Inductor	1.35 mH	9.45	Link
L5	Inductor	0.68 mH	7.95	Link
R2	Resistor	15.0 Ω	0.99	Link
C7	Capacitor	6.0 μF	8.45	Link
		Estimated Total	**81.25**	

3 Crossover Design

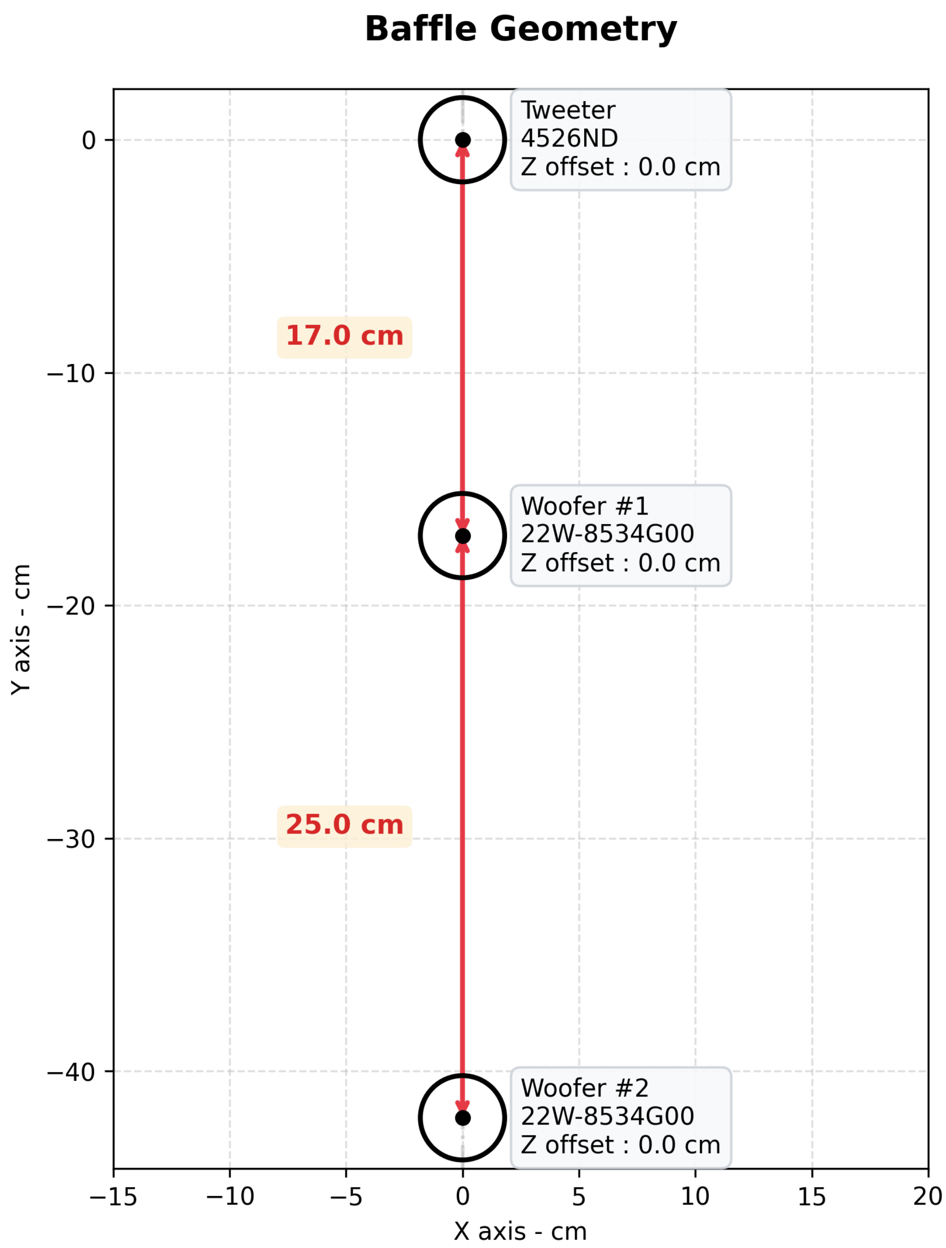
This section provides the electrical schematic required to build the crossover and achieve the acoustic results presented earlier. If you need help understanding this diagram, please refer to the beginner guide.

3.1 Electrical Schematic



3.2 Baffle Geometry

All figures are computed using the following baffle layout. We recommend using this layout for optimal acoustic performance, but feel free to experiment with different placements if you have specific constraints or preferences. Just keep in mind that drastically changing the geometry may affect the frequency response and directivity of your speakers due to phase shifts.


Baffle Geometry
Tweeter
4526ND
Z offset : 0.0 cm
17.0 cm
Woofer #1
22W-8534G00
Z offset : 0.0 cm
25.0 cm
Woofer #2
22W-8534G00
Z offset : 0.0 cm
Y axis - cm
X axis - cm
0
−10
−20
−30
−40
−15
−10
−5
0
5
10
15
20

THANK YOU

FOR YOUR SUPPORT

I hope this guide helps you build the crossover for your DIY audio project.

If you have any questions, run into issues, or just want to share a picture of your finished build, please feel free to reach out via Etsy messages!

Happy building,

Audio440

www.etsy.com/shop/Audio440